stat443_10.21

2026-10-21

1. Data Loading and Cleaning

```
library(readxl)
library(dplyr)
library(ggplot2)

data <- read_excel("~/Desktop/Group3Data.xlsx")
```

```
clean_data <- data %>%
  filter(
    !is.na(shares), !is.na(num_imgs), !is.na(num_videos),
    !is.na(num_hrefs), shares >= 0, num_imgs >= 0,
    num_videos >= 0, num_hrefs >= 0
  )

tech_clean <- clean_data %>%
  filter(data_channel_is_tech == 1)

nrow(tech_clean)
```

```
## [1] 7346
```

2. Initial Regression Data

```
reg_data <- tech_clean %>%
  select(shares, num_imgs, num_videos, num_hrefs,
         n_tokens_content, global_sentiment_polarity) %>%
  filter(complete.cases(.), shares >= 0) %>%
  mutate(log_shares = log1p(shares))

nrow(reg_data)
```

```
## [1] 7346
```

```
summary(reg_data)
```

```
##      shares          num_imgs        num_videos        num_hrefs     
##  Min.   :    36   Min.   : 0.000   Min.   : 0.0000   Min.   :  0.000  
##  1st Qu.:  1100   1st Qu.: 1.000   1st Qu.: 0.0000   1st Qu.:  5.000  
##  Median :  1700   Median : 1.000   Median : 0.0000   Median :  7.000  
##  Mean   :  3072   Mean   : 4.435   Mean   : 0.4472   Mean   :  9.417  
##  3rd Qu.:  3000   3rd Qu.: 6.000   3rd Qu.: 1.0000   3rd Qu.: 11.000  
##  Max.   :663600   Max.   :65.000   Max.   :73.0000   Max.   :120.000  
##  n_tokens_content global_sentiment_polarity   log_shares    
##  Min.   :   0.0   Min.   :-0.33912          Min.   : 3.611  
##  1st Qu.: 256.0   1st Qu.: 0.09713          1st Qu.: 7.004  
##  Median : 405.0   Median : 0.14379          Median : 7.439  
##  Mean   : 571.6   Mean   : 0.14577          Mean   : 7.582  
##  3rd Qu.: 728.0   3rd Qu.: 0.19102          3rd Qu.: 8.007  
##  Max.   :5530.0   Max.   : 0.65000          Max.   :13.405
```

3. Model 1: Basic Linear Regression

```
model1 <- lm(log_shares ~ num_imgs + num_videos + num_hrefs,
             data = reg_data)
summary(model1)
```

```
##
## Call:
## lm(formula = log_shares ~ num_imgs + num_videos + num_hrefs,
##     data = reg_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.0479 -0.5437 -0.1412  0.4347  5.6105
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.470896   0.014385 519.362  < 2e-16 ***
## num_imgs    -0.001061   0.001441  -0.736    0.462
## num_videos   0.024852   0.006072   4.093 4.31e-05 ***
## num_hrefs    0.011119   0.001188   9.362  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8058 on 7342 degrees of freedom
## Multiple R-squared:  0.01577,    Adjusted R-squared:  0.01536
## F-statistic:  39.2 on 3 and 7342 DF,  p-value: < 2.2e-16
```

4. Model 2: Extended Linear Regression

```
model2 <- lm(
  log_shares ~ num_imgs + num_videos + num_hrefs +
    n_tokens_content + global_sentiment_polarity,
  data = reg_data
)
summary(model2)
```

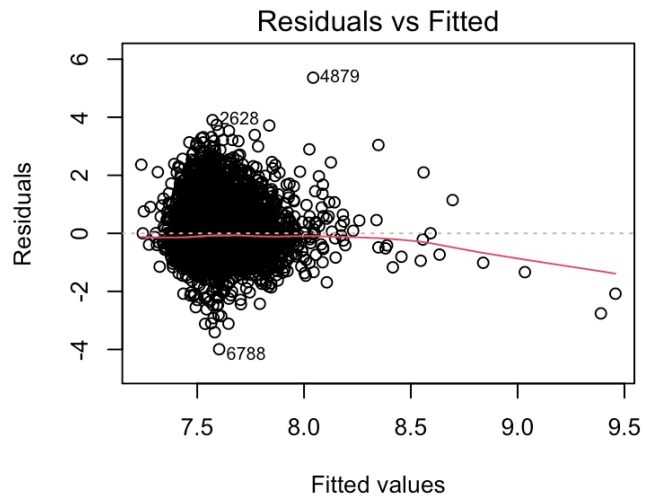
```
##
## Call:
## lm(formula = log_shares ~ num_imgs + num_videos + num_hrefs +
##     n_tokens_content + global_sentiment_polarity, data = reg_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.9916 -0.5398 -0.1401  0.4304  5.3628
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 7.518e+00  2.290e-02 328.248  < 2e-16 ***
## num_imgs                   -3.754e-03  1.572e-03  -2.388 0.016968 *
## num_videos                  2.180e-02  6.087e-03   3.581 0.000344 ***
## num_hrefs                   8.798e-03  1.316e-03   6.685 2.48e-11 ***
## n_tokens_content            1.199e-04  2.474e-05   4.845 1.29e-06 ***
## global_sentiment_polarity -5.551e-01  1.224e-01  -4.534 5.89e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8034 on 7340 degrees of freedom
## Multiple R-squared:  0.02191,    Adjusted R-squared:  0.02125
## F-statistic: 32.89 on 5 and 7340 DF,  p-value: < 2.2e-16
```

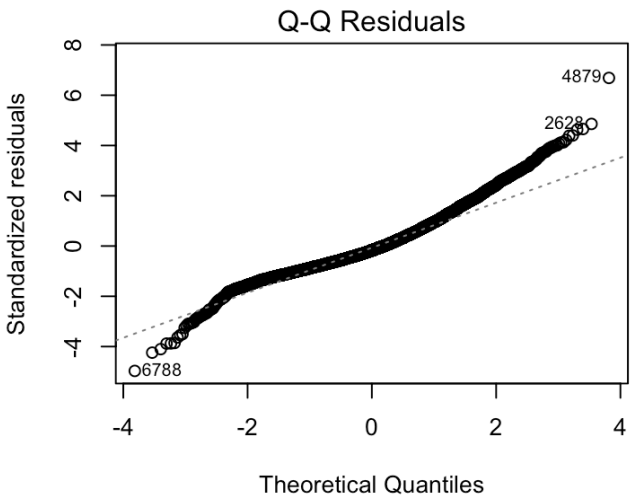
```
data.frame(
  Model = c("Basic Model", "Extended Model"),
  Adjusted_R2 = c(summary(model1)$adj.r.squared,
                  summary(model2)$adj.r.squared),
  AIC = c(AIC(model1), AIC(model2))
)
```

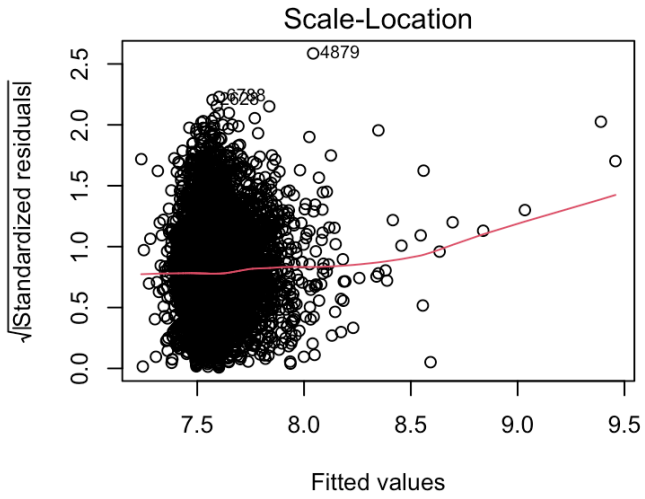
```
##            Model Adjusted_R2      AIC
## 1    Basic Model  0.01536414 17680.10
## 2 Extended Model  0.02124689 17638.08
```

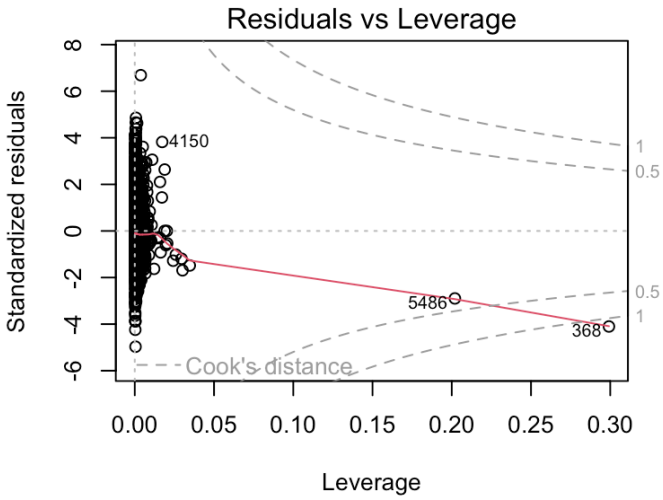
Model 2 Diagnostics

```
par(mfrow = c(2, 2))
plot(model2)
```










```
par(mfrow = c(1, 1))
```

5. Prepare Weekday Variables

```
reg_data3 <- tech_clean %>%
  mutate(
    weekday = case_when(
      weekday_is_monday == 1 ~ "Monday",
      weekday_is_tuesday == 1 ~ "Tuesday",
      weekday_is_wednesday == 1 ~ "Wednesday",
      weekday_is_thursday == 1 ~ "Thursday",
      weekday_is_friday == 1 ~ "Friday",
      weekday_is_saturday == 1 ~ "Saturday",
      weekday_is_sunday == 1 ~ "Sunday",
      TRUE ~ NA_character_
    ),
    weekday = factor(
      weekday,
      levels = c("Monday", "Tuesday", "Wednesday", "Thursday",
                 "Friday", "Saturday", "Sunday")
    )
  ) %>%
  filter(
    complete.cases(shares, num_imgs, num_videos, num_hrefs,
                   n_tokens_content, global_sentiment_polarity, weekday),
    shares >= 0
  )

nrow(reg_data3)
```

```
## [1] 7346
```

6. Model 3: Original Shares and Weekday

```
model3 <- lm(
  shares ~ num_imgs + num_videos + num_hrefs +
    n_tokens_content + global_sentiment_polarity + weekday,
  data = reg_data3
)
summary(model3)
```

```
##
## Call:
## lm(formula = shares ~ num_imgs + num_videos + num_hrefs + n_tokens_content +
##     global_sentiment_polarity + weekday, data = reg_data3)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -14918  -1835  -1140     82 656488
##
## Coefficients:
##                               Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 2261.2495   343.8402   6.576 5.15e-11 ***
## num_imgs                     -49.8641    17.6252  -2.829 0.004680 **
## num_videos                   141.6274    68.0722   2.081 0.037510 *
## num_hrefs                     66.4373    14.8093   4.486 7.36e-06 ***
## n_tokens_content               1.0191     0.2769   3.680 0.000235 ***
## global_sentiment_polarity -3429.4052  1370.5252  -2.502 0.012362 *
## weekdayTuesday               118.5170   347.2151   0.341 0.732860
## weekdayWednesday             584.0021   350.6690   1.665 0.095877 .
## weekdayThursday              -12.5204   357.2343  -0.035 0.972042
## weekdayFriday                266.5017   383.9537   0.694 0.487642
## weekdaySaturday              620.4031   469.5646   1.321 0.186466
## weekdaySunday                855.5980   520.0865   1.645 0.099991 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8980 on 7334 degrees of freedom
## Multiple R-squared:  0.01118,    Adjusted R-squared:  0.009698
## F-statistic: 7.539 on 11 and 7334 DF,  p-value: 4.812e-13
```

```
model3_no_weekday <- lm(
  shares ~ num_imgs + num_videos + num_hrefs +
    n_tokens_content + global_sentiment_polarity,
  data = reg_data3
)

anova(model3_no_weekday, model3)
```

```
## Analysis of Variance Table
##
## Model 1: shares ~ num_imgs + num_videos + num_hrefs + n_tokens_content +
##     global_sentiment_polarity
## Model 2: shares ~ num_imgs + num_videos + num_hrefs + n_tokens_content +
##     global_sentiment_polarity + weekday
##   Res.Df        RSS Df Sum of Sq      F Pr(>F)
## 1   7340 5.9204e+11
## 2   7334 5.9148e+11  6 564747582 1.1671 0.3207
```

7. Model 3: Log-Transformed Shares and

Weekday

```
model3_log <- lm(
  log1p(shares) ~ num_imgs + num_videos + num_hrefs +
    n_tokens_content + global_sentiment_polarity + weekday,
  data = reg_data3
)

model_log_no_weekday <- lm(
  log1p(shares) ~ num_imgs + num_videos + num_hrefs +
    n_tokens_content + global_sentiment_polarity,
  data = reg_data3
)

anova(model_log_no_weekday, model3_log)
```

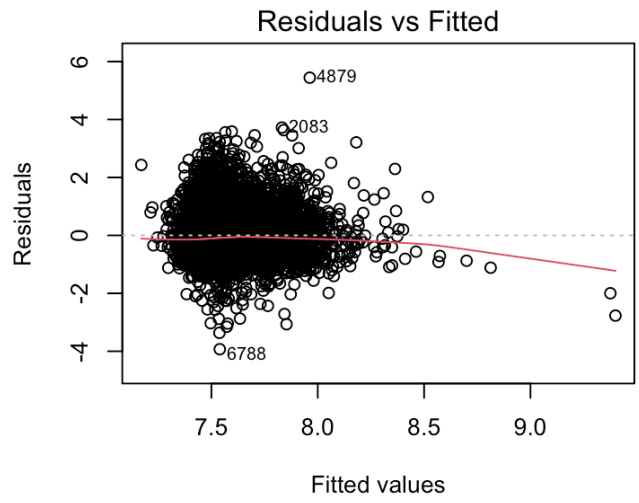
```
## Analysis of Variance Table
##
## Model 1: log1p(shares) ~ num_imgs + num_videos + num_hrefs + n_tokens_content +
##     global_sentiment_polarity
## Model 2: log1p(shares) ~ num_imgs + num_videos + num_hrefs + n_tokens_content +
##     global_sentiment_polarity + weekday
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1   7340 4737.1
## 2   7334 4657.4  6    79.664 20.908 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

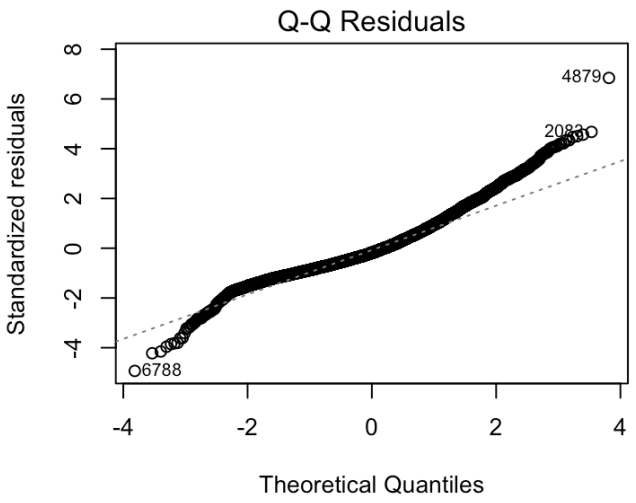
```
summary(model3_log)
```

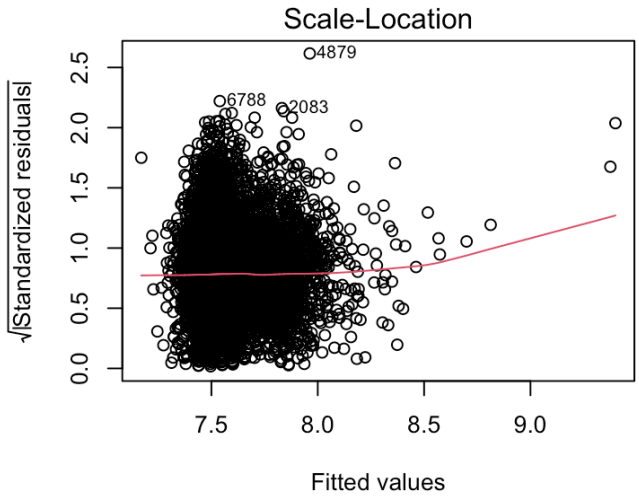
```
##
## Call:
## lm(formula = log1p(shares) ~ num_imgs + num_videos + num_hrefs +
##     n_tokens_content + global_sentiment_polarity + weekday, data = reg_data3)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.9284 -0.5392 -0.1472  0.4201  5.4432
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 7.487e+00  3.051e-02 245.378  < 2e-16 ***
## num_imgs                   -3.108e-03  1.564e-03  -1.987 0.046958 *
## num_videos                  2.262e-02  6.040e-03   3.744 0.000182 ***
## num_hrefs                   7.275e-03  1.314e-03   5.536 3.20e-08 ***
## n_tokens_content            1.179e-04  2.457e-05   4.799 1.63e-06 ***
## global_sentiment_polarity -5.667e-01  1.216e-01  -4.660 3.22e-06 ***
## weekdayTuesday              2.260e-03  3.081e-02   0.073 0.941528
## weekdayWednesday           -3.099e-03  3.112e-02  -0.100 0.920675
## weekdayThursday            -1.807e-02  3.170e-02  -0.570 0.568725
## weekdayFriday               7.178e-02  3.407e-02   2.107 0.035169 *
## weekdaySaturday             3.045e-01  4.167e-02   7.308 3.01e-13 ***
## weekdaySunday               3.252e-01  4.615e-02   7.046 2.01e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7969 on 7334 degrees of freedom
## Multiple R-squared:  0.03836,    Adjusted R-squared:  0.03692
## F-statistic:  26.6 on 11 and 7334 DF,  p-value: < 2.2e-16
```

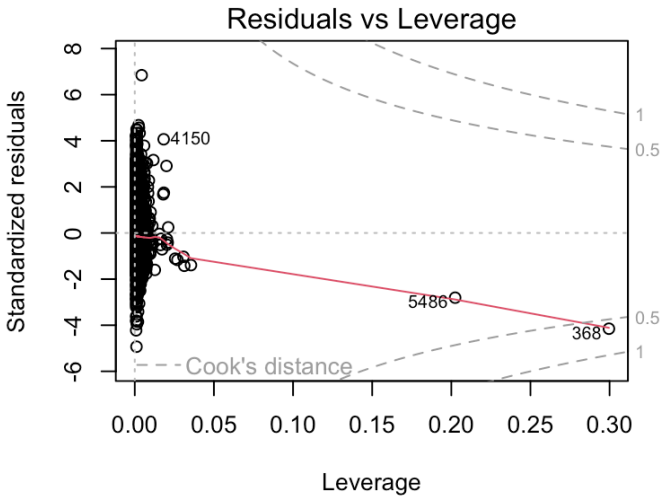
Model 3 Log Diagnostics

```
par(mfrow = c(2, 2))
plot(model3_log)
```










```
par(mfrow = c(1, 1))
```

8. Model 4: Weekday vs Weekend

```
reg_data3 <- reg_data3 %>%
  mutate(
    weekend = factor(
      ifelse(weekday %in% c("Saturday", "Sunday"),
             "Weekend", "Weekday"),
      levels = c("Weekday", "Weekend")
    )
  )

model_weekend <- lm(
  log1p(shares) ~ num_imgs + num_videos + num_hrefs +
    n_tokens_content + global_sentiment_polarity + weekend,
  data = reg_data3
)
summary(model_weekend)
```

```
##
## Call:
## lm(formula = log1p(shares) ~ num_imgs + num_videos + num_hrefs +
##     n_tokens_content + global_sentiment_polarity + weekend, data = reg_data3)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.9400 -0.5394 -0.1475  0.4201  5.4335
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 7.494e+00  2.284e-02 328.170  < 2e-16 ***
## num_imgs                   -3.175e-03  1.561e-03  -2.034 0.041983 *
## num_videos                  2.259e-02  6.040e-03   3.740 0.000185 ***
## num_hrefs                   7.384e-03  1.312e-03   5.627 1.90e-08 ***
## n_tokens_content            1.165e-04  2.455e-05   4.747 2.10e-06 ***
## global_sentiment_polarity -5.674e-01  1.215e-01  -4.671 3.05e-06 ***
## weekendWeekend              3.060e-01  2.828e-02  10.819  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7971 on 7339 degrees of freedom
## Multiple R-squared:  0.03727,    Adjusted R-squared:  0.03648
## F-statistic: 47.35 on 6 and 7339 DF,  p-value: < 2.2e-16
```

9. Simpler Model

```
model_simple <- lm(
  log1p(shares) ~ num_imgs + num_videos + num_hrefs + weekend,
  data = reg_data3
)
summary(model_simple)
```

```
##
## Call:
## lm(formula = log1p(shares) ~ num_imgs + num_videos + num_hrefs +
##     weekend, data = reg_data3)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.9961 -0.5427 -0.1465  0.4166  5.6766
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)     7.4441332  0.0144861 513.882  < 2e-16 ***
## num_imgs       -0.0005737  0.0014304  -0.401    0.688
## num_videos      0.0255594  0.0060253   4.242 2.24e-05 ***
## num_hrefs       0.0096173  0.0011865   8.106 6.10e-16 ***
## weekendWeekend  0.3064697  0.0283667  10.804  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7995 on 7341 degrees of freedom
## Multiple R-squared:  0.03117,    Adjusted R-squared:  0.03064
## F-statistic: 59.05 on 4 and 7341 DF,  p-value: < 2.2e-16
```

10. Full vs Simple Model Comparison

```
AIC(model_weekend, model_simple)
```

```
##               df      AIC
## model_weekend  8 17523.85
## model_simple   6 17566.22
```

```
summary(model_weekend)$adj.r.squared
```

```
## [1] 0.03648005
```

```
summary(model_simple)$adj.r.squared
```

```
## [1] 0.030643
```

```
anova(model_simple, model_weekend)
```

```
## Analysis of Variance Table
##
## Model 1: log1p(shares) ~ num_imgs + num_videos + num_hrefs + weekend
## Model 2: log1p(shares) ~ num_imgs + num_videos + num_hrefs + n_tokens_content +
##     global_sentiment_polarity + weekend
##   Res.Df    RSS Df Sum of Sq      F   Pr(>F)
## 1   7341 4692.2
## 2   7339 4662.7  2    29.525 23.236 8.72e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

11. Findings

1. Weekends seem to matter.Technology articles published on weekends were associated with about 36% more shares than those published on weekdays.
2. Videos and links seem to be useful predictors.Both variables were significant in the full and simpler models.However, the results for images are less clear.The number of images was significant in the full model but not in the simpler model.
3. The full model performed better.It had a higher adjusted R-squared and a lower AIC.The ANOVA test also showed that adding article length and sentiment significantly improved the model fit.
4. The R-squared is low.The full model only explained about 3.7% of the variation in shares.
5. For now, I think we could keep the full model.The weekend finding might be worth highlighting.